**Project Design Phase-I**
**Solution Architecture**

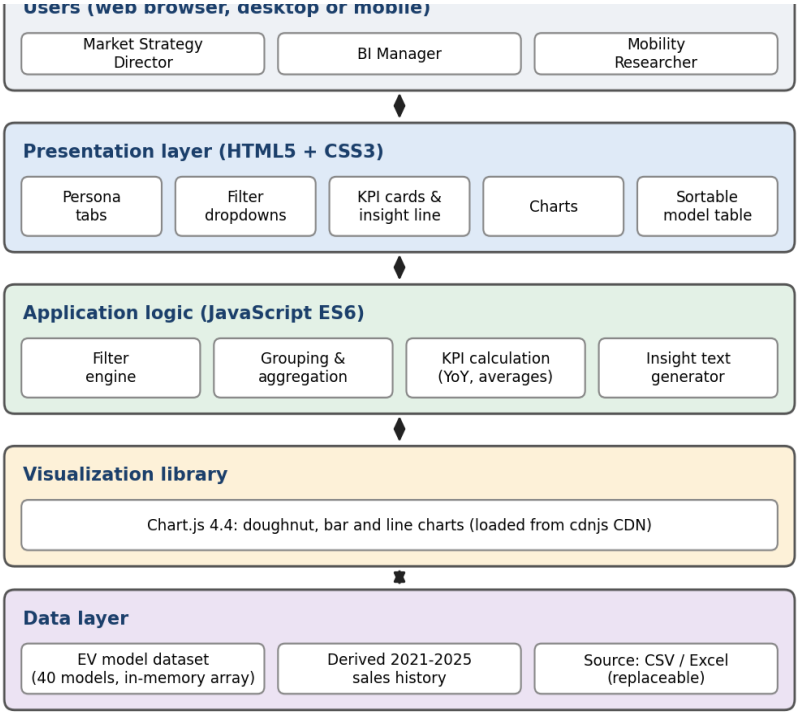
| | |
|---|---|
| Date | 19 September 2026 |
| Team ID | PNT2022TMIDxxxxxx |
| Project Name | EV Market Intelligence Dashboard |
| Maximum Marks | 4 Marks |

## Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

## EV Market Intelligence Dashboard – Solution Architecture Diagram:




**Component Description Table**

| Component Name | Description / Role in Architecture | Technology Used |
|---|---|---|
| Users / Clients | Strategy, BI and Research users open the dashboard in a web browser on desktop or mobile. | Modern web browser |
| User Interface (Front-End) | Persona tabs, filter dropdowns, KPI cards, insight banner and sortable model table; responsive with light and dark themes. | HTML5, CSS3 |
| Application Logic | Applies filters, groups and sums data, calculates KPIs (YoY growth, weighted range and rating, shares) and builds the insight text. | JavaScript (ES6) |
| Data Visualization | Draws doughnut, bar and line charts that update with filters and theme. | Chart.js 4.4 |
| Data Layer | EV model dataset (40 models: sales, battery, range, price, rating, country) plus derived 2021-2025 sales history. | In-memory JavaScript array; CSV / Excel source |
| Hosting & Delivery | Serves the single HTML page and loads Chart.js from a CDN. | Static web hosting, cdnjs CDN |